William Roche

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EDUCATION

New York University — **New York, NY**
Bachelor of Science in Computer Science, Minor in Mathematics — *September 2025 - May 2027*
GPA: 3.946 | Dean's List

The College of New Jersey — **Ewing, NJ**
Major in Computer Science, Minor in Mathematics — *September 2023 - May 2025 (Transferred)*
GPA: 3.92 | Dean's List | Upsilon Pi Epsilon Honor Society

EXPERIENCE

Natixis CIB | *Software Engineering Intern* — New York, NY | June 2026 – Aug 2026
- Automated a previously manual quarterly MPID reconciliation between internal systems and external FINRA/NASDAQ data via a Levenshtein-based name matching ETL pipeline, flagging gaps in ~7% of records for compliance remediation
- Modernized legacy reporting across credit risk, intraday liquidity, and treasury by migrating reports to SQL/Power BI
- Eliminated manual report distribution by building Python scripts to generate and email Excel reports every working day

NYU CQIP | *Quantum Computing Research Intern* — New York, NY | June 2025 – Aug 2025
- Fabricated a vertical graphene Josephson junction to enable ballistic supercurrent transport across a ~0.4nm channel
- Engineered channel geometry around graphene's coherence length, since Cooper pairs decouple beyond a monolayer
- Used photolithography, sputtering, and PDMS transfer; characterized with AFM and cryogenic resistance measurement

Freelance | *Social Media Creator & Growth Consultant* — Remote | Jan 2024 – May 2025
- Grew a personal brand to over 3M views, 22K followers, and $25,000 in revenue via algorithm-driven growth strategies
- Mentored 7 small-scale creators on TikTok Live, driving a 38% average revenue increase through heightened viewership

The College of New Jersey | *Software Engineering Intern* — Ewing, NJ | Sept 2023 – May 2024
- Produced a cross-platform PWA to improve campus resource access for 7,000+ students by modifying TCNJ Library App
- Aimed to improve student efficiency by tracking availability of 27 study rooms via MySQL/PHP and motion sensor data

PROJECTS

RiskRadar | **CMU TartanHacks 2026** | *Pandas, Numpy, BeautifulSoup4, Dedalus MCP, React, TypeScript, Node.js*
- Created an internal risk assessment tool for banks, leveraging public records and social media to profile loan applicants
- Gathered data from 400+ platforms in 15 minutes through a parallelized OSINT pipeline running across 6 async waves

LemonLink | **Morgan Stanley Code To Give 2026** | *Next.js, FastAPI, Supabase (PostgreSQL), Mapbox GL*
- Built an admin analytics dashboard to track volunteer activity, event participation, and platform-wide engagement metrics
- Developed a context-aware AI assistant that personalized support responses based on each user's profile and event data

Gainsight | **Portfolio Tax Tracker** | *Java, Spring Boot, React, PostgreSQL, Groq API*
- Devised a FIFO tax engine with wash sale detection, rolling disallowed losses into a short/long-term tax liability estimate
- Architected 5 custom exception classes mapped to HTTP status codes via a global handler, backed by a JUnit test suite

Custom Linux Shell | **Linux Shell** | *C*
- Designed a Unix-like shell supporting command history, piping, I/O redirection, process execution, and Linux commands
- Implemented process management, directory management, and IPC with system calls (fork, execvp, pipe, chdir)

SKILLS

Programming Languages: Python, C++, Java, SQL, C, JavaScript, TypeScript, PHP, Ruby
Frameworks & Libraries: React, Node.js, Pandas, NumPy, FastAPI, Next.js, Spring Boot, Tailwind CSS
Tools & Platforms: Git, Linux, Docker, AWS, REST APIs, PostgreSQL, CI/CD, MySQL, Vercel, Supabase, Jira, Alteryx

RELEVANT COURSEWORK

Algorithms, Operating Systems, Computer Architecture, Probability and Statistics, Machine Learning, Software Engineering

INTERESTS

Cooking, Freshwater Fishing, Hiking, Lord of the Rings, Minecraft, Shrimpkeeping, Soccer, Stock Trading, Weightlifting